

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that:

The initial outgoing request from the browser to the DNS server was sent via UDP to resolve the domain name `www.yummyrecipesforme.com`. The source IP `192.51.100.15` sent a DNS query to destination IP `203.0.113.2` on port 53, requesting an A record for the domain.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: "udp port 53 unreachable"

The port noted in the error message is used for:

Changing Domain name into IP address

The most likely issue is:

the initial outgoing request from the browser to the DNS server was sent via UDP to resolve the domain name `www.yummyrecipesforme.com`. The source IP `192.51.100.15` sent a DNS query to destination IP `203.0.113.2` on port 53, requesting an A record for the domain.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: 13:24.192571

Explain how the IT team became aware of the incident: Several customers reported that they were not able to access the website and saw error message "destination port unreachable"

Explain the actions taken by the IT department to investigate the incident:

They load the dcp dump, the analyzer showed that when you send UDP Packet to the DNS server, you receive ICMP packet containing the error message 'udp port 53 unreachable'

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

The client (`192.51.100.15`) sent a UDP DNS query to the DNS server (`203.0.113.2`) on port 53.

- The DNS server responded with an ICMP error message: udp port 53 unreachable.

- This indicates that no DNS service was listening on port 53 of the server, causing the query to fail.
- The browser was unable to resolve the domain name to an IP address, which prevented the HTTPS request from being sent to the web server.
- The error occurred consistently across three separate ICMP responses, confirming the issue was not transient.

Note a likely cause of the incident: A likely cause is that the DNS service on the server was stopped, crashed, or was intentionally disabled, or a firewall/security group is blocking UDP traffic on port 53. Without the DNS service active, the server could not process domain resolution requests, resulting in the ICMP "port unreachable" responses and making the website inaccessible to all users.